A First-Order Primal-Dual Algorithm for Convex Problems with Applications to Imaging

Paper by Antonin Chambolle and Thomas Pock (2011)

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Introduction

Paper: Antonin Chambolle, Thomas Pock. *A first-order primal-dual algorithm for convex problems with applications to imaging*. 2010. ⟨hal-00490826⟩

Problem:

- Variational methods are essential for ill-posed inverse imaging problems.
- While total variation regularization preserves sharp discontinuities (edges), it is difficult to minimize due to the non-smoothness of the functional.
- The authors introduce a flexible first-order primal-dual algorithm specifically designed for structured, non-smooth convex optimization problems.
- Unlike prior schemes that either require expensive linear system solves per iteration or stall at slow subgradient rates, this method achieves an optimal $O(1/N)$ convergence rate for general non-smooth problems.
- Particularly, they achieve $O(1/N^2)$ convergence when either the primal or dual objective is uniformly convex, and linear convergence $O(1/e^N)$ when both are uniformly convex.

Optimization Problem

$$\min_{x \in X} \; F(Kx) + G(x)$$

such that:

- X, Y: finite-dimensional real vector spaces.
- $K : X \to Y$ is continuous, linear; $\|K\| = \max\{\|Kx\| : x \in X, \|x\| \leq 1\} < \infty$.
- $G : X \to [0, \infty)$ and $F : Y \to [0, \infty)$ are proper, convex, l.s.c. functions.

By **Fenchel-Legendre duality**, $F(Kx) = \max_{y \in Y} \langle Kx, y \rangle - F^*(y)$ [1], converting the primal problem into the **saddle-point problem**:

$$\min_{x \in X} \max_{y \in Y} \langle Kx, y \rangle + G(x) - F^*(y) \quad (1)$$

Assuming:

- There exists saddle point $(\hat{x}, \hat{y})$ satisfying:

$$K\hat{x} \in \partial F^*(\hat{y}), \quad -K^*\hat{y} \in \partial G(\hat{x})$$

- F, G are "simple": resolvents $(I + \tau \partial F)^{-1}(y) = \arg\min_x \left\{ \frac{\|x-y\|^2}{2\tau} + F(x) \right\}$ and $(I + \tau \partial G)^{-1}$ have closed-form solutions or can be solved efficiently.

[1] For an operator A, A^* denotes its complex conjugate.

Main Algorithm

Initialization: Choose $\tau, \sigma > 0$ with $\tau\sigma\|K\|^2 < 1$, $\theta \in [0,1]$, $(x^0, y^0) \in X \times Y$, and set $\bar{x}^0 = x^0$.

For each iteration $n \geq 0$: (2)

1. **Dual Proximal Ascent Step (y^{n+1}):**
   - Applying proximal ascent and completing the square yields:

$$y^{n+1} = (I + \sigma\partial F^*)^{-1}(y^n + \sigma K\bar{x}^n) = \arg\min_{y \in Y}\left\{\frac{\|y - (y^n + \sigma K\bar{x}^n)\|^2}{2\sigma} + F^*(y)\right\}$$

2. **Primal Proximal Descent Step (x^{n+1}):**
   - Similarly,

$$x^{n+1} = (I + \tau\partial G)^{-1}\left(x^n - \tau K^* y^{n+1}\right) = \arg\min_{x \in X}\left\{\frac{\|x - (x^n - \tau K^* y^{n+1})\|^2}{2\tau} + G(x)\right\}$$

   Both updates are obtained by setting the subgradient of their respective objectives to zero:

$$\frac{y^n - y^{n+1}}{\sigma} + K\bar{x}^n \in \partial F^*(y^{n+1})$$

$$\frac{x^n - x^{n+1}}{\tau} - K^* y^{n+1} \in \partial G(x^{n+1})$$

3. **Linear Extrapolation ($\bar{x}^{n+1}$):**

$$\bar{x}^{n+1} = x^{n+1} + \theta(x^{n+1} - x^n)$$

   Acts as an approximate extragradient step along the primal update direction, anticipating the next position, helping stabilize the primal-dual iterations.

Central Theorem

Let $L = \|K\|$ and assume problem (1) has a saddle-point $(\hat{x}, \hat{y})$. Choose $\theta = 1$, $\tau\sigma L^2 < 1$, and let $(x_n, \bar{x}_n, y_n)$ be defined by (2). Then:

(a) For any n,

$$\frac{\|y^n - \hat{y}\|^2}{2\sigma} + \frac{\|x^n - \hat{x}\|^2}{2\tau} \leq C\left(\frac{\|y^0 - \hat{y}\|^2}{2\sigma} + \frac{\|x^0 - \hat{x}\|^2}{2\tau}\right) \quad (3),$$

where the constant $C \leq (1 - \tau\sigma L^2)^{-1}$.

(b) If we let

$$x_N = \frac{1}{N}\sum_{n=1}^{N} x^n, \qquad y_N = \frac{1}{N}\sum_{n=1}^{N} y^n,$$

for any bounded $B_1 \times B_2 \subset X \times Y$, the restricted gap has the following bound:

$$\mathcal{G}_{B_1 \times B_2}(x_N, y_N) \leq \frac{D(B_1, B_2)}{N}, \quad (3)$$

where

$$\mathcal{G}_{B_1 \times B_2}(x, y) = \max_{y' \in B_2}\left\{\langle y', Kx\rangle - F^*(y') + G(x)\right\} - \min_{x' \in B_1}\left\{\langle y, Kx'\rangle - F^*(y) + G(x')\right\}.$$

and

$$D(B_1, B_2) = \sup_{(x,y) \in B_1 \times B_2}\left\{\frac{\|x - x^0\|^2}{2\tau} + \frac{\|y - y^0\|^2}{2\sigma}\right\}.$$

Moreover, the weak cluster points of (x_N, y_N) are saddle-points of (2);

(c) If the dimension of the spaces X and Y is finite, then there exists a saddle-point (x^*, y^*) such that

$$x^n \to x^* \qquad \text{and} \qquad y^n \to y^*.$$

Proof Idea: From Descent to Convergence

1. Optimality Condition

$$\frac{x^n - x^{n+1}}{\tau} - K^* y^{n+1} \in \partial G(x^{n+1}),$$
$$\frac{y^n - y^{n+1}}{\sigma} + K\bar{x}^n \in \partial F^*(y^{n+1}).$$

2. Telescoping and Ergodic Averages

Summing the subgradient inequalities over N iterations and using the step-size condition $\tau\sigma\|K\|^2 < 1$, we obtain:

$$\sum_{n=0}^{N-1} \left[\mathcal{L}(x^{n+1}, y) - \mathcal{L}(x, y^{n+1})\right]$$
$$\leq E_0(x, y) - E_N(x, y) \leq E_0(x, y).$$

where,

- $\mathcal{L}(x, y) = G(x) + \langle Kx, y\rangle - F^*(y)$ is the Lagrangian.
- $E_n(x, y)$ is the weighted squared distance from the n-th iterate to (x, y):

$$E_n(x, y) = \frac{\|x^n - x\|^2}{2\tau} + \frac{\|y^n - y\|^2}{2\sigma}.$$

By convexity, for ergodic averages

$$x_N = \frac{1}{N}\sum_{n=1}^{N} x^n, \qquad y_N = \frac{1}{N}\sum_{n=1}^{N} y^n,$$

we obtain

$$\boxed{G_{B_1 \times B_2}(x_N, y_N) \leq \frac{D(B_1, B_2)}{N}}$$

Thus, the ergodic primal-dual gap converges at a rate of $O(1/N)$.

4. Iterate Convergence

- (3) ensures that the iterates are bounded. Since X and Y are finite-dimensional, there exists a convergent subsequence.
- Using the vanishing increments and the proximal optimality conditions, the limit of this subsequence is shown to be a saddle point.
- Fejér monotonicity then establishes convergence of the entire sequence to that saddle point.

Implementation: Total Variation L^1 Denoising

We consider the **TV-L^1** model, a non-smooth optimization problem.

TV-L^1 Model

$$\min_u \|\nabla u\|_1 + \lambda \|u - g\|_1$$

$$\approx$$

$$\min_u \max_p \{-\langle u, \operatorname{div} p\rangle + \lambda \|u - g\|_1 - \delta_P(p)\}$$

$$\lambda = 1.5$$

where

$u \in \mathbb{R}^{M\times N}$	:	Restored image,
$g \in \mathbb{R}^{M\times N}$	:	Noisy input image,
λ	:	Regularization parameter,
$p \in \mathbb{R}^{2\times M\times N}$	:	Dual variable,
$\operatorname{div} p$	:	Divergence of p,
$\delta_P(p)$	:	Indicator function of set P.

$$P = \{\, p \in \mathbb{R}^{2\times M\times N} : \|p_{ij}\|_2 \le 1, \quad \forall\, i = 1,\dots,M,\ j = 1,\dots,N \,\}.$$

Method

The primal-dual algorithm (2) [**ALG1**]

$$\tau = 0.02, \qquad \tau\sigma L^2 = 1, \qquad \theta = 1 \qquad L = \|\nabla\|$$

Experimental Setup

We take an image with

$$M = 500, \qquad N = 375,$$

where the noisy input g is corrupted with 25% salt-and-pepper noise.

Ground-truth method

A well-performing method run over $\sim 10^5$ iters.

Stopping Criterion

$$\frac{E_n - E^*}{E^*} < \epsilon \qquad \epsilon \in \{10^{-4}, 10^{-5}\}$$

where E_n and E^* are the primal energy of the current iterate u_n and of the reference.

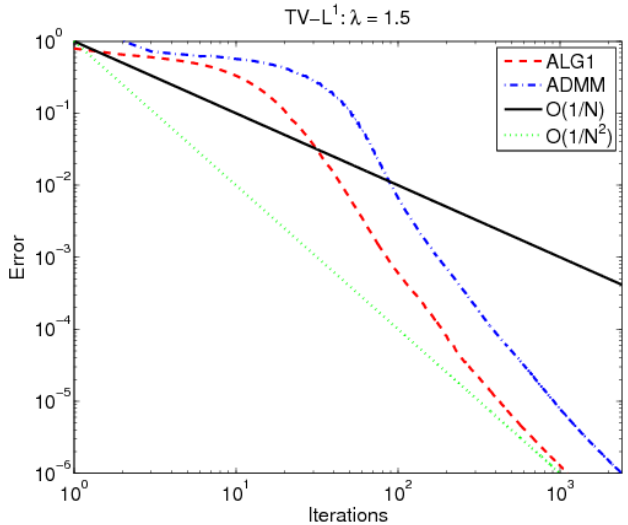
Implementation

All algorithms are implemented in Matlab and executed on a 2.66 GHz CPU, running a 64 Bit Linux system.

[1] The experiment uses the boundary case $\tau\sigma L^2 = 1$. Remark 1 notes that while control of the estimate for y^N is lost, an estimate for x^N remains. Full convergence is not guaranteed.

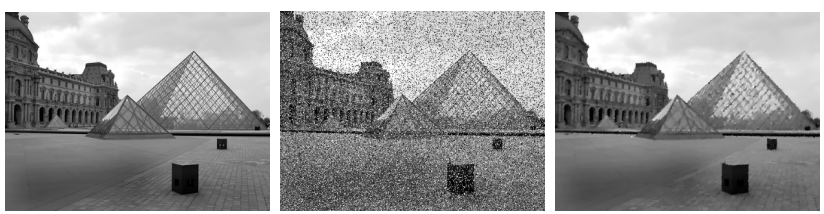
Experiment Results: $TV\text{-}L^1$ Denoising

Convergence



Exhibits faster convergence than other methods.

Denoising Results



(a) Clean (b) Noisy (S&P) (c) ALG1

Removes image noise while preserving edges.

Quantitative Performance

Method	Iterations $\varepsilon = 10^{-4}$	Iterations $\varepsilon = 10^{-5}$	∇/div ($\times 10^5$)	CPU (s) ($\times 10^5$)
ALG1	**187**	**421**	**842**	**36.0**
ADMM	385	916	—	80.0
EGRAD	2462	8736	34,944	1360.0
NEST	2406	15538	—	1387.0

ADMM: Alternating Direction Method; EGRAD: Extra-Gradient; NEST: Nesterov.

Key Takeaways

- ALG1 is 2.2× **faster** than ADMM.
- The observed convergence is consistent with the theoretical $O(1/N)$ rate, with the empirical tail approaching $O(1/N^2)$.
- Removes noise robustly while preserving image quality.

New Application: Color Denoising in 3D Point Clouds

Problem Setup

A 3D point cloud consists of spatial points with associated RGB color values. We construct a spatial graph and independently denoise each color channel while preserving spatial discontinuities.

1. Graph Construction
Given a point cloud $P = \{p_1, \ldots, p_N\}$, $p_i \in \mathbb{R}^3$:

- Each 3D point represents a graph vertex.
- Connect each vertex to its k nearest spatial neighbours.

2. Edge Weights
Assign weights based on spatial proximity:

$$w_{ij} = \exp\left(-\frac{\|p_i - p_j\|_2^2}{h^2}\right),$$

where h is the median positive edge length.

3. Graph Gradient
Let u_i^c denote the color intensity at point p_i for channel $c \in \{R, G, B\}$. For each edge:

$$(Ku^c)_{ij} = \sqrt{w_{ij}}(u_j^c - u_i^c).$$

Graph-TV Denoising

4. Graph Total Variation
Aggregate weighted intensity differences across neighbouring points:

$$TV_G(u^c) = \sum_i \sqrt{\sum_{j\sim i} w_{ij}(u_j^c - u_i^c)^2}.$$

5. Optimization Model
For each color channel independently:

$$\min_{u^c} \; TV_G(u^c) + \frac{\lambda}{2}\|u^c - f^c\|_2^2$$

- f^c, u^c: noisy and denoised intensities.
- λ: balances smoothness and data fidelity.

6. Chambolle–Pock Solver

- Formulate graph-TV denoising as a primal-dual problem.
- Apply Chambolle–Pock using the graph gradient operator K.

Results: Color Denoising in 3D Point Clouds

- We consider Keenan's model of the cow, Spot.[2]. 8000 points are sampled from its mesh with Gaussian noise added to colour channels of each point.
- We implement the Chambole-Pock method to denoise this noisy point cloud under both the $TV - L^1$ model and ROF (L_2 norm in TV) and measure the Mean Squared Error (MSE) wrt the original model.

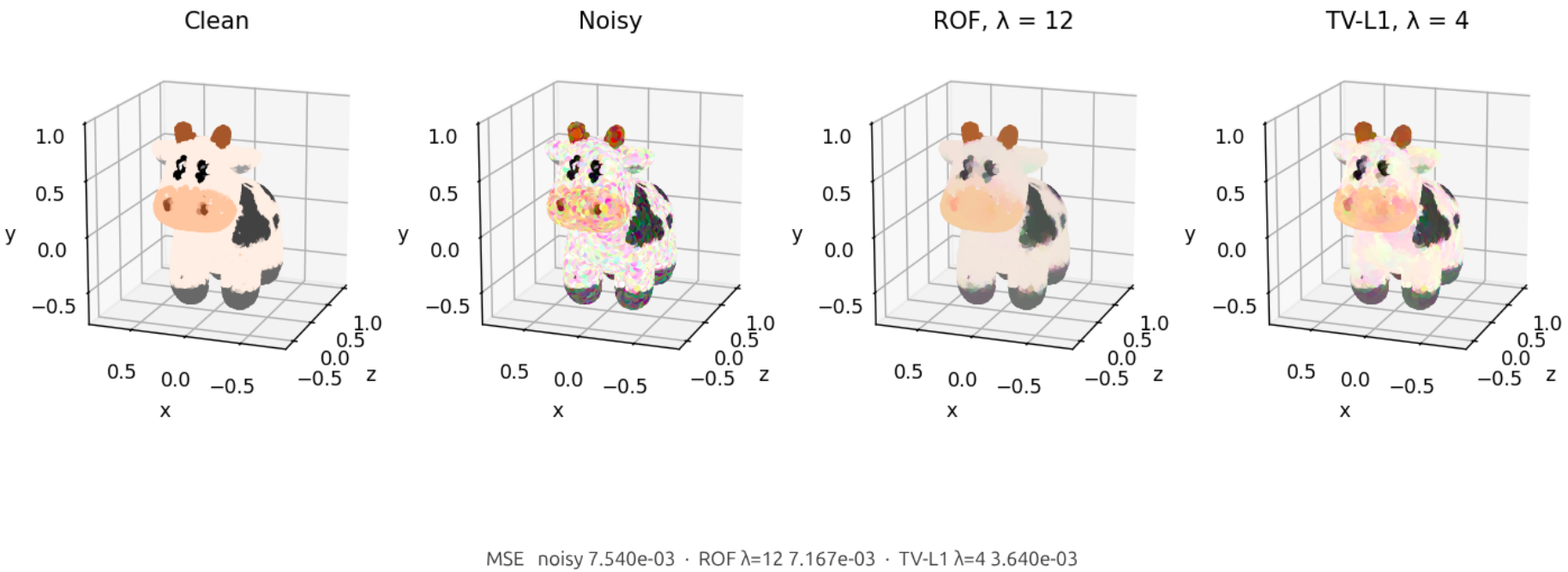


Figure: Color denoising of Spot under *ROF* (MSE drop by 5%) and TVL^1 (MSE drop by 50%).

[2] https://www.cs.cmu.edu/~kmcrane/Projects/ModelRepository/

Failure Cases, Assumptions and Limitations

Theoretical Limitations

- Requires a convex saddle-point formulation with proper, convex, lower-semicontinuous objective functions.
- Convergence requires step sizes satisfying $\tau\sigma\|K\|^2 < 1$; equality is not covered by the standard convergence guarantee.

Computational Limitations

- The step-size constraint can lead to slow convergence when $\|K\|$ is large, restricting the allowable primal and dual step sizes.
- Computational cost can be high when the proximal operators do not admit efficient closed-form solutions.

Empirical Limitations

- **Staircasing:** First-order TV regularization can produce artificial piecewise-constant regions, particularly in smoothly varying image regions.
- **Edge blurring:** Denoising can attenuate or blur image edges, especially under strong regularization.
- **Noise–detail trade-off:** Increasing regularization suppresses more noise but can also remove fine structures and image details. The appropriate balance depends on the noise level and image characteristics and needs tuning λ.

Key Takeaways

1. Handling Non-Smooth Convex Optimization

Total Variation (TV) regularization is widely used in image denoising and many applications, but it leads to non-smooth, non-differentiable optimization problems.

- Chambolle–Pock handles such problems using proximal operators and subgradients, without requiring the objective functions to be differentiable.
- Its primal-dual splitting approach exploits the individual structures of F and G, making complex optimization problems easier to solve.

2. Faster Convergence Through Stronger Convexity

The convexity properties of the objective functions can be exploited to achieve faster convergence rates.

- $O(1/N)$ convergence rate for the ergodic primal-dual gap.
- Uniform convexity of one component enables $O(1/N^2)$ convergence.
- Uniform convexity of both components enables linear convergence $O(1/e^N)$.

3. A Flexible Framework for Diverse Applications

The Chambolle–Pock framework can be adapted to a wide range of imaging problems by choosing suitable optimization formulations and proximal operators.

- The paper demonstrates its applications in image denoising, image deconvolution, image inpainting, motion estimation, and image segmentation.
- We can extend the 2D problems to 3D point-cloud denoising, using a k-nearest-neighbour graph and graph based total variation.